

Lecture: System Administration and Maintenance

I. Introduction

In any organization that uses computers and networks, there are people responsible for making sure that everything runs smoothly — these people are called **System Administrators** (or **SysAdmins**).

They maintain the systems, fix problems, install software, and make sure everything is secure and working properly.

System Administration and Maintenance is about keeping the **computers, servers, and network systems** running efficiently and securely.

II. Definition of System Administration

System Administration is the process of managing and maintaining computer systems, servers, and networks to ensure they operate reliably, efficiently, and securely.

III. Roles and Responsibilities of a System Administrator

1. User Management

- Create and manage user accounts.
- Set permissions and access rights.
- Reset passwords and manage authentication.

2. System Installation and Configuration

- Install operating systems (Windows, Linux, macOS).
- Configure hardware and software settings.
- Ensure compatibility between devices.

3. Maintenance and Updates

- Apply system updates and patches.
- Perform regular maintenance to prevent issues.

- Remove unused or harmful software.

4. Backup and Recovery

- Regularly back up important data.
- Prepare recovery plans for system failure or data loss.

5. Security Management

- Protect systems from viruses and cyber threats.
- Install firewalls and antivirus software.
- Monitor user activity for suspicious behavior.

6. Network Management

- Maintain network connections and routers/switches.
- Assign IP addresses and configure connectivity.
- Troubleshoot network issues.

7. System Monitoring

- Monitor system performance (CPU, memory, storage).
- Detect and fix problems early before they cause downtime.

8. Documentation

- Keep records of system configurations and updates.
- Document troubleshooting steps and system policies.

IV. Types of System Maintenance

1. Preventive Maintenance

- Regular checkups to avoid future problems.
- Example: Cleaning dust from computer components or updating software.

2. Corrective Maintenance

- Fixing issues after they occur.
- Example: Replacing a damaged hard drive or reinstalling corrupted software.

3. Adaptive Maintenance

- Updating systems to work with new technologies.
- Example: Upgrading operating systems to support new hardware.

4. Perfective Maintenance

- Improving performance and adding new features.
- Example: Enhancing security features or improving system speed.

V. Tools Used in System Administration

- **Command-Line Interface (CLI)** – For configuration and control.
- **Remote Desktop / SSH** – For accessing systems remotely.
- **Monitoring Tools** – e.g., Nagios, Zabbix, SolarWinds.
- **Backup Software** – e.g., Acronis, Veeam.
- **Antivirus / Firewall Tools** – For system protection.
- **Task Scheduler / Cron Jobs** – To automate maintenance tasks.

VI. Importance of System Administration and Maintenance

- Ensures **continuous operation** of IT services.
- Protects **data integrity and security**.
- Reduces **downtime** and **productivity loss**.
- Keeps **systems updated** and **efficient**.
- Provides **support** to users and ensures smooth operations.

VII. Common Problems and Troubleshooting

Problem	Possible Cause	Solution
Slow computer	Too many background apps	Disable unnecessary startup programs

No internet	Network misconfiguration	Check router, cables, and IP settings
Virus infection	Unsecured browsing	Run antivirus scan and update definitions
Login issues	Incorrect password	Reset password or check user permissions

VIII. Best Practices for System Administrators

- Always perform **regular backups**.
- Keep systems and software **updated**.
- Use **strong passwords** and multi-factor authentication.
- Maintain **proper documentation**.
- Monitor logs and system performance.
- Educate users on **cybersecurity awareness**.
- Test recovery plans regularly.

IX. Summary

- **System Administration** ensures systems and networks work properly.
- **Maintenance** keeps them running efficiently and securely.
- A good System Administrator is **organized, proactive, and security-conscious**.

X. Discussion Questions

1. Why is preventive maintenance better than corrective maintenance?
2. What could happen if a company does not back up its data regularly?
3. How can a system administrator protect users from cybersecurity threats?

Activity

